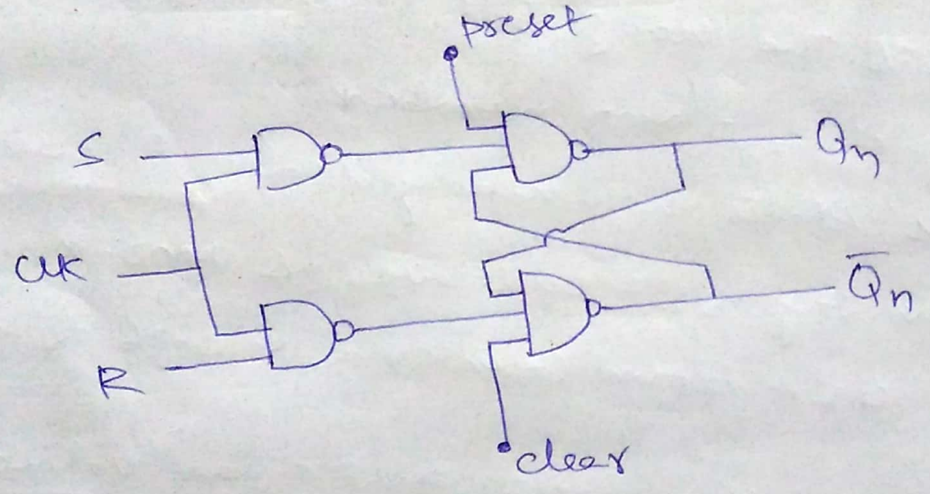
preset and clear inputs

→ These are called direct inputs or overriding inputs or asynchronous inputs.

→ These are called overriding inputs because independent of flipflop inputs when these inputs are present, o/p of the flip flop changes.



Here, when preset $= 0 \Rightarrow Q_n = 1$

when clear $= 0 \Rightarrow \bar{Q}_n = 1 \Rightarrow Q_n = 0$

whatever be the clock and inputs of the flipflop, ~~they~~ when present $= 0$ my Q_n value is 1. Similarly, when clear $= 0$ my Q_n value is 0.

→ If preset and clear are equal to 1, then flipflop performs normally and it is independent of preset and clear inputs.

(Note: Roll Numbers should not be written)

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Modulus of the counter & counting to particular value

⇒ 2 bit ripple counter is called MOD-4 or modulus 4 counter.

⇒ 3 bit ripple counter is called MOD-8 or modulus 8 counter.

⇒ If n is the no. of bits then MOD number = 2^n.

no. of states = MOD number.

⇒ Modulus states how many states it having.

⇒ So we can say 3 bit up counter as MOD-8 up counter.

Similarly, 4 bit down counter as MOD-16 down counter.

⇒ **How to stop counting upto particular value**

Ex: Design MOD-6 counter using MOD-8 counter.

Now we have to design MOD-8 counter. we can design MOD-8 means 3 bit counter. from MOD-8 counter either up/down counter. i.e it must we have to design MOD-6 counter. have 6 states and Max. count is 5.

In MOD-8 counter,

no. of states = 8

no. of bits = 3.

no. of FF's = 3

Max Count = 8 - 1 = 7.

where as in MOD-6 counter,

no. of states = 6

no. of bits = 3

no. of FF's = 3

Max. count = 6 - 1 = 5.

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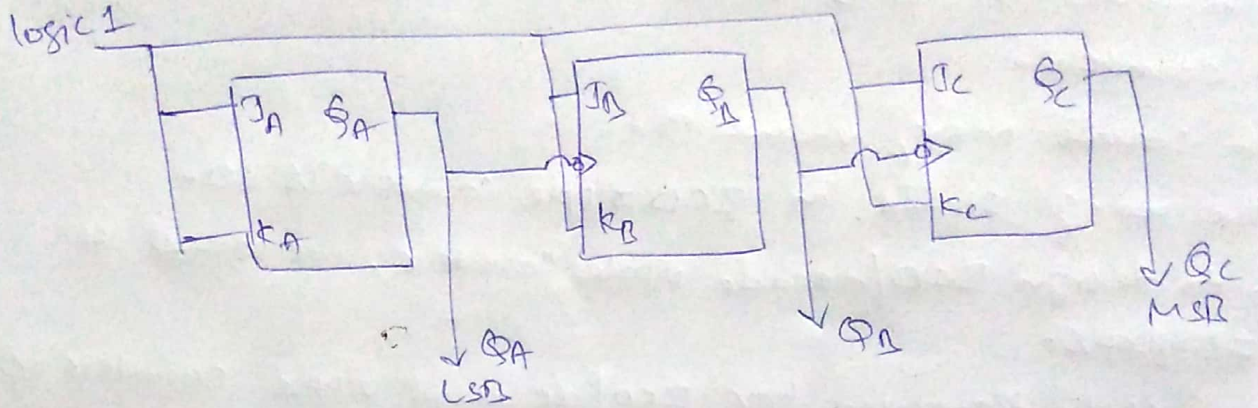
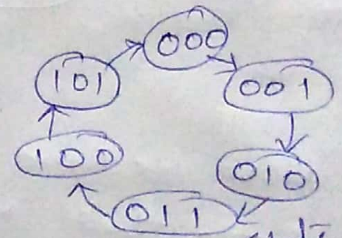
first we have to design MOD-8 counter. Here i'm considering up counter.

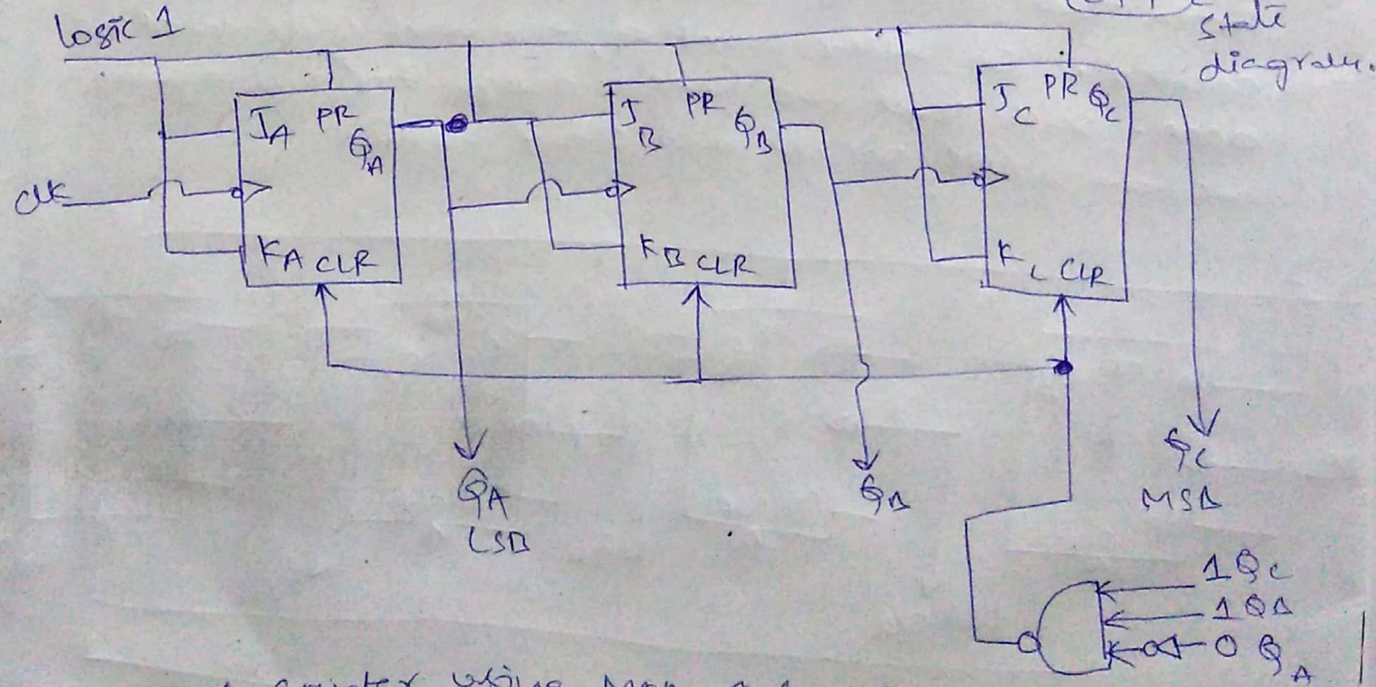


From this Mod-8 counter, we have to design Mod-6 counter. For this, i use preset and clear inputs. i give preset input to logic 1. So the circuit is independent of preset input. But Mod-8 counter counts upto 7. But here in Mod-6 counter, max. count is 5 only. So when ~~its counts~~ Mod-8 counter counts 5, after that next count is 0, then it acts as Mod-6 counter. for that, we give clear i/p = 0 when Mod8 counter counts 6.

6 = 1 1 0



state diagram.



MOD-6 counter using MOD-8 counter.

Decade (BCD) ripple counter

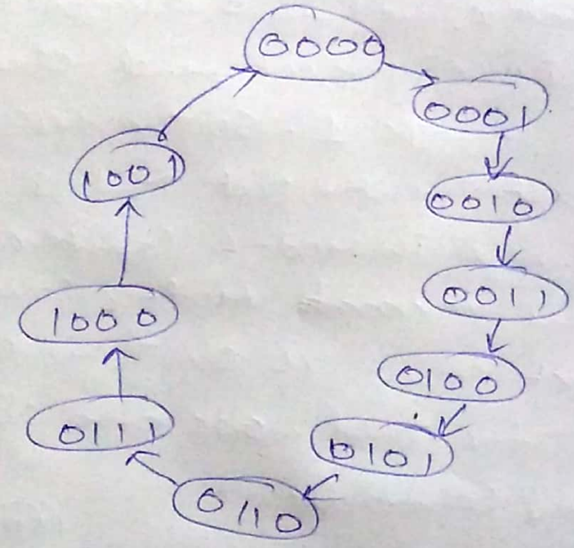
⇒ Decade (or) BCD ripple counter means Mod-10 counter.

it counts Max. value 9.

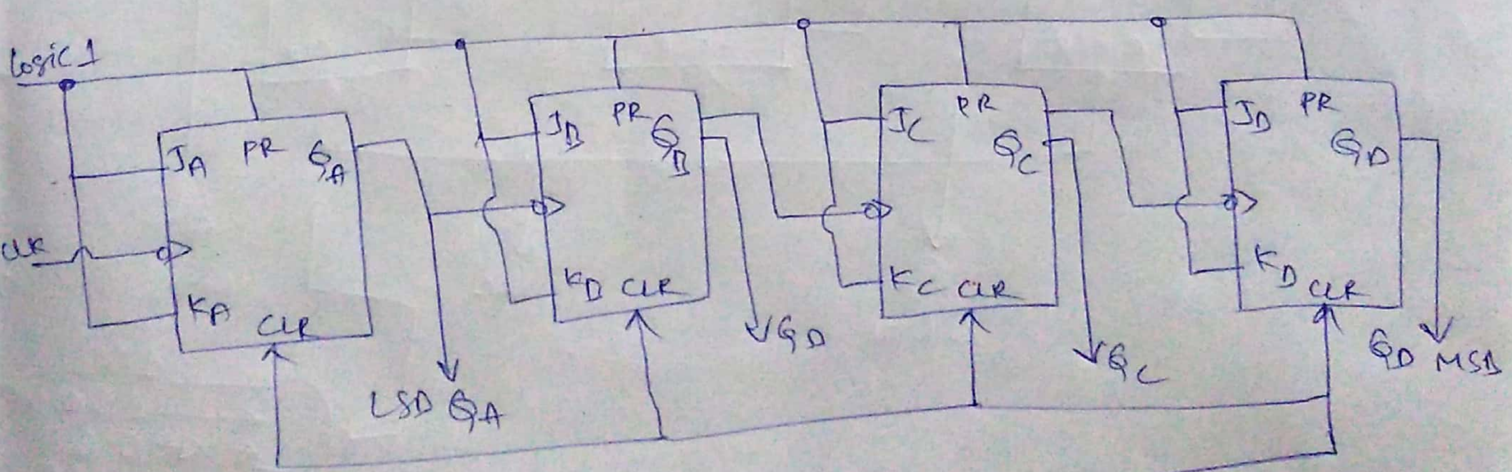
the no. of states in BCD ripple counter is 10.

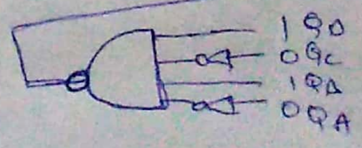
So to design BCD/decade ripple counter, we need 4 Flipflops.

the state diagram for BCD/decimal ripple counter is,



Now we have to design $2^4 = 16$ i.e MOD 16 counter.
From that MOD-16 counter, we develop Mod 10|BCD counter. So, after count 9, the next state must be 0000. such that we have to design. for this again we have to use preset and clear inputs.





MOD-10 counter using MOD-16 counter.

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